Chaos Language Algorithm: Testing Summary

Chaos Language Algorithm Project

2026-07-10

Abstract

This document summarizes all testing performed on the Chaos Language Algorithm (CLA) prototype as of 2026-07-10. The pure-Python `chaoslang` library has been tested on five deterministic dynamical control systems (logistic map, Lorenz-63, Rossler, Mackey-Glass, Lorenz-96) across dimensions 1 through 24, plus a 1024-step × 20-dimensional OmegaSim-scale smoke run. Every recorded experiment preserves exact symbolic reconstruction. The prototype learns chunk grammars under an approximate MDL objective, with a bounded suffix-trie miner and Jensen-Shannon category proposal path. No non-chaotic controls, shuffled surrogates, held-out evaluation, matched brute-force comparison, or actual OmegaSim traces have been tested yet.

1 Systems Tested

System	Dimension	Integration
Logistic map	1	Discrete iteration (r=4.0, x_0=0.123)
Lorenz-63	3	RK4 (σ=10, ρ=28, β=8/3, dt=0.01)
Rossler	3	RK4 (a=0.2, b=0.2, c=5.7, dt=0.05)
Mackey-Glass	1	Euler with 17-step delay (dt=0.1)
Lorenz-96	5–24	RK4 (F=8.0, dt=0.01)

Table 1: Dynamical control systems.

2 Experiment Matrix

2.1 256-symbol baseline suite

All runs: 256 retained symbols, 64 discarded, 4 bins/axis, 8 greedy iterations, seed 0.

2.2 1024-symbol OmegaSim-scale smoke

3 Unit Tests

The repository test suite grows across sprints:

Test coverage includes: exact expansion after edits, chunk identity, category identity with $M[v]$ member retention, frame generalization, MDL rejection of rare-symbol categories, rule pruning, determinism, compound-symbol mining equivalence, bounded trie node counts, and M1 symbolization round-trip for all five attractor systems.

System	Dim	Unique	Rules	Score	Exact
Mackey-Glass	1	4	6	44.4	yes
Logistic map	1	4	8	113.2	yes
Lorenz-63	3	21	8	187.2	yes
Rossler	3	24	8	189.2	yes
Lorenz-96	5	56	8	234.2	yes
Lorenz-96	8	78	8	229.2	yes
Lorenz-96	12	107	6	241.4	yes
Lorenz-96	16	121	3	248.7	yes
Lorenz-96	20	127	6	251.4	yes
Lorenz-96	24	148	5	251.5	yes

Table 2: 256-symbol baseline runs. Score = approximate MDL proxy units.

Parameter	Value
System	Lorenz-96
Dimension	20
Steps	1024
Discard	256
Bins	4
Iterations	8
Miner	suffix_trie
Category method	JS (threshold 0.1)
Seed	0
Commit	b344aa3 (+ dirty CLI mod)
Unique symbols	510
Rules learned	8
Score	1021.2
Fit wall time	2.55 s
Exact reconstruction	yes

Table 3: 1024-step $\times$ 20D suffix-trie smoke run.

4 Key Findings

1. **Exact reconstruction holds universally.** All 11 recorded experiments (10 baselines + 1 OmegaSim-scale smoke) verify that the learned grammar expands to the original symbol stream.

2. **Compression degrades with dimension.** At dim 1–3, scores range 44–189 (0.17–0.74 units/symbol). At dim 16–24, scores plateau near 249–252 (0.97–0.98 units/symbol), approaching raw parse cost. This reflects the 4^D compound-symbol explosion under direct M1 binning, not absence of dynamical structure.

3. **Mackey-Glass is most compressible.** Only 4 unique symbols and score 44.4, consistent with its quasi-periodic behavior at these parameters.

4. **OmegaSim-scale smoke succeeds.** The 1024 $\times$ 20D run completed in 2.55 s with exact reconstruction, 510 unique symbols, and 8 chunk rules. No category edit was accepted, so

Date	Milestone	Tests
2026-07-03	Sprint 1: symbolic-string MVP	19
2026-07-04	M1 controls + JS seam	26
2026-07-09	Suffix-trie miner + JS integration	69

Table 4: Unit-test growth. All tests passing at each checkpoint.

the JS path was exercised but not validated as beneficial on this run.

5. **Suffix-trie miner is functional.** The bounded trie miner handles high-cardinality 20D streams without OOM. However, no matched brute-force comparison exists to quantify speedup or memory savings.

5 Limitations

- No non-chaotic controls, shuffled surrogates, or periodic baselines: sensitivity/specificity for chaos detection is unknown.
- No held-out evaluation: M1 bounds are inferred from the full trajectory.
- Score is a heuristic proxy, not a calibrated information code.
- Single-run per parameter setting: no confidence intervals or perturbation sweeps.
- Peak memory not recorded for any run.
- No matched brute-force vs. trie benchmark.
- 256-symbol records lack raw stdout, environment captures, and commit identifiers.
- No actual OmegaSim trace tested: "OmegaSim-scale" refers to stream length and dimensionality only.
- No category edit was accepted in the 1024 × 20D run; JS clustering benefit is unvalidated.

6 Repository State

- GitHub: `bgoertzel-sing/chaos-language-algorithm` (public)
- Local: `projects/chaos-language-algorithm/repos/chaoslang`
- HEAD: `b344aa3` (local `main`, 2 commits ahead of `origin/main`)
- Branch `agent/suffix-trie-miner` merged to local `main`
- Test command: `PYTHONPATH=src python3 -m unittest discover -s tests -v`

7 Next Steps

1. Matched brute-force vs. suffix-trie benchmark with peak RSS capture.
2. Non-chaotic and surrogate controls for classification validation.
3. Held-out and multi-seed evaluation protocol.
4. Dimension-reduction symbolization beyond direct 4-bin compound M1.
5. McBride-derivative pre-ranking (M-MD1) and joint chunk/category proposals (M-MD2).
6. Per-iteration MDL trajectory logging (M-MD3).
7. Run on actual OmegaSim traces after export schema is fixed.